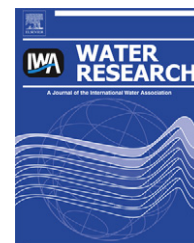


Available at www.sciencedirect.comjournal homepage: www.elsevier.com/locate/watres

Kinetic assessment and modeling of an ozonation step for full-scale municipal wastewater treatment: Micropollutant oxidation, by-product formation and disinfection

Saskia G. Zimmermann^{a,b,1}, Mathias Wittenwiler^{a,b}, Juliane Hollender^{a,2},
Martin Krauss^{a,3,7}, Christoph Ort^{a,6,4}, Hansruedi Siegrist^{a,5}, Urs von Gunten^{a,b,*}

^a Eawag, Swiss Federal Institute of Aquatic Science and Technology, P.O. Box 611, 8600 Dübendorf, Switzerland

^b Institute of Biogeochemistry and Pollutant Dynamics, ETH Zürich, 8092 Zürich, Switzerland

ARTICLE INFO

Article history:

Received 23 March 2010

Received in revised form

28 June 2010

Accepted 28 July 2010

Available online 5 August 2010

Keywords:

Micropollutant oxidation

Rct

Modeling

Full-scale ozonation

Oxidation by-products

Disinfection

ABSTRACT

The kinetics of oxidation and disinfection processes during ozonation in a full-scale reactor treating secondary wastewater effluent were investigated for seven ozone doses ranging from 0.21 to 1.24 g O₃ g⁻¹ dissolved organic carbon (DOC). Substances reacting fast with ozone, such as diclofenac or carbamazepine ($k_{p,O_3} > 10^4 \text{ M}^{-1} \text{ s}^{-1}$), were eliminated within the gas bubble column, except for the lowest ozone dose of 0.21 g O₃ g⁻¹ DOC. For this low dose, this could be attributed to short-circuiting within the reactor. Substances with lower ozone reactivity ($k_{p,O_3} < 10^4 \text{ M}^{-1} \text{ s}^{-1}$) were only fully eliminated for higher ozone doses.

The predictions of micropollutant oxidation based on coupling reactor hydraulics with ozone chemistry and reaction kinetics were up to a factor of 2.5 higher than full-scale measurements. Monte Carlo simulations showed that the observed differences were higher than model uncertainties. The overestimation of micropollutant oxidation was attributed to a protection of micropollutants from ozone attack by the interaction with aquatic colloids. Laboratory-scale batch experiments using wastewater from the same full-scale treatment plant could predict the oxidation of slowly-reacting micropollutants on the full-scale level within a factor of 1.5. The Rct value, the experimentally determined ratio of the concentrations of hydroxyl radicals and ozone, was identified as a major contribution to this difference.

An increase in the formation of bromate, a potential human carcinogen, was observed with increasing ozone doses. The final concentration for the highest ozone dose of 1.24 g O₃ g⁻¹ DOC was 7.5 µg L⁻¹, which is below the drinking water standard of 10 µg L⁻¹. N-Nitrosodimethylamine (NDMA) formation of up to 15 ng L⁻¹ was observed in the first compartment of the reactor, followed by a slight elimination during sand filtration. Assimilable organic carbon (AOC) increased up to 740 µg AOC L⁻¹, with no clear trend when correlated to the ozone dose, and decreased by up to 50% during post-sand filtration. The

* Corresponding author. Tel.: +41 44 823 5270; fax: +41 44 823 5210.

E-mail addresses: saskia.zimmermann@eawag.ch (S.G. Zimmermann), mathias.wittenwiler@gmail.com (M. Wittenwiler), juliane.hollender@eawag.ch (J. Hollender), martin.krauss@ufz.de (M. Krauss), c.ort@awmc.uq.edu.au (C. Ort), hansruedi.siegrist@eawag.ch (H. Siegrist), vongunten@eawag.ch (U. von Gunten).

¹ Tel.: +41 44 823 5083; fax: +41 44 823 5028.

² Tel.: +41 44 823 5493; fax: +41 44 823 5893.

³ Tel.: +49 341 235 1530; fax: +49 341 235 45 1530.

⁴ Tel.: +61 7 3345 4730; fax: +61 7 3365 4726.

⁵ Tel.: +41 44 823 5054; fax: +41 44 823 5389.

⁶ Present address: The University of Queensland, Advanced Water Management Centre (AWMC), Qld 4072, Australia.

⁷ Present address: UFZ – Helmholtz Centre for Environmental Research, Permoserstr. 15, 04318 Leipzig, Germany.

0043-1354/\$ – see front matter © 2010 Elsevier Ltd. All rights reserved.

doi:10.1016/j.watres.2010.07.080

disinfection capacity of the ozone reactor was assessed to be 1–4.5 log units in terms of total cell counts (TCC) and 0.5 to 2.5 log units for *Escherichia coli* (*E. coli*). Regrowth of up to 2.5 log units during sand filtration was observed for TCC while no regrowth occurred for *E. coli*. *E. coli* inactivation could not be accurately predicted by the model approach, most likely due to shielding of *E. coli* by flocs.

© 2010 Elsevier Ltd. All rights reserved.

1. Introduction

In recent years, mitigation strategies to lower micropollutant loads from wastewater treatment plant (WWTP) effluents to receiving waters have been discussed intensively. Ozonation is a promising technique to upgrade WWTPs with a tertiary treatment step because it has been demonstrated to be efficient for the oxidation of pharmaceuticals and other micropollutants in drinking water (Ternes et al., 2002; Westerhoff et al., 2005) and wastewater (Hollender et al., 2009; Huber et al., 2005; Nakada et al., 2007; Ternes et al., 2003; Wert et al., 2009a,b). Oxidation does not usually result in complete mineralization, but in the formation of transformation products, which in many cases have a much lower biological activity than the parent compounds (Dodd et al., 2009; Huber et al., 2004; Lee et al., 2008). In addition to the formation of transformation products, other undesired by-products can be formed such as bromate, a potential human carcinogen, formed during ozonation of bromide-containing waters by a complicated mechanism involving both ozone and hydroxyl radicals (von Gunten, 2003a). N-Nitrosodimethylamine (NDMA) formation was reported during ozonation of drinking water (Schmidt and Brauch, 2008) and wastewater (Hollender et al., 2009; Yang et al., 2009), while N-Nitrosomorpholine (NMOR) formation occurred during ozonation of lake water (Zhao et al., 2008). Other organic by-products are formed from the oxidative breakdown of complex dissolved organic matter (DOM), which are usually readily biodegradable and can be measured as e.g. assimilable organic carbon (AOC) (Hammes et al., 2006; von Gunten, 2003a). AOC is an important water quality parameter for the biological stability of drinking water and can influence river water quality after discharge of ozonated wastewater. For all of these known oxidation by-products, no information is available on the extent and kinetics of their formation during ozonation of secondary wastewater effluent. Concerning the toxicity of ozonated secondary wastewater effluent, a decrease in specific and unspecific toxicity after ozonation was demonstrated by an *in vitro* test battery using different toxicological endpoints (Escher et al., 2009), while *in vivo* tests with rainbow trout suggested an increased toxicity that disappeared after sand filtration (Stalter et al., 2010). This can be explained by the formation of potentially toxic by-products from DOM oxidation, such as aldehydes, ketones and other oxygen-rich compounds (von Gunten, 2003a), which might not be enriched during solid phase extraction necessary for the *in vitro* test battery. These oxygen-rich compounds are typically removed during sand filtration (Hammes et al., 2006).

Besides its role as an oxidant, ozone is mainly used in drinking water production because it is an effective disinfectant for viruses, bacteria and protozoa (von Gunten, 2003a). In

the present study, the main focus is on micropollutant oxidation, however, disinfection is an additional benefit especially in reference to the amended bathing water directive of the European Union (2006/7/EC). Both objectives can be achieved by ozonation at a reasonable and manageable cost as demonstrated by Hollender et al. (2009).

Prediction of the oxidation and disinfection capacity of a full-scale ozonation step is of practical and scientific interest to ensure optimum performance. The design of the reactor as well as site-specific wastewater composition are important parameters influencing oxidation and disinfection performance of ozonation steps. von Gunten et al. (1997, 1999) developed a concept to predict the behavior of oxidants and micropollutants in an ozone reactor treating lake water for drinking water purposes. This concept is based on the coupling of reactor hydraulics with ozone chemistry and reaction kinetics, and has been successfully applied for the simulation of ozone reactors treating spring (Boller et al., 2000) and river water (Gallard et al., 2003).

The objectives of the present study were to assess the oxidation and disinfection capacity of a full-scale ozonation step by quantitatively describing the kinetics of the following processes occurring during ozonation of secondary wastewater effluent: (i) micropollutant oxidation, (ii) oxidation by-product formation, (iii) AOC formation, and (iv) disinfection. (v) Finally, two different approaches to predict ozone reactor performance in terms of micropollutant oxidation were tested: (a) a model based on the concept of von Gunten et al. (1997, 1999) which could also be used for disinfection prediction, and (b) spiked laboratory-scale batch experiments using secondary effluent from the same full-scale WWTP.

2. Materials and methods

2.1. Standards and reagents

All chemicals and solvents were of analytical purity ($\geq 95\%$) and purchased from various suppliers. The micropollutants analyzed in the sampling campaigns are compiled in Table S1 of the Supplementary Information (SI).

2.2. Ozone reactor

A detailed description of WWTP Wüeri in Regensdorf, Switzerland, including rapid sand filtration as final biological treatment step after ozonation was given by Hollender et al. (2009). A mixing chamber for final flocculation was retrofitted to a full-scale six compartments ozone reactor with a volume of 38 m³ (Fig. 1). A gaseous ozone/oxygen mixture was added in counter-current mode through a series of disc

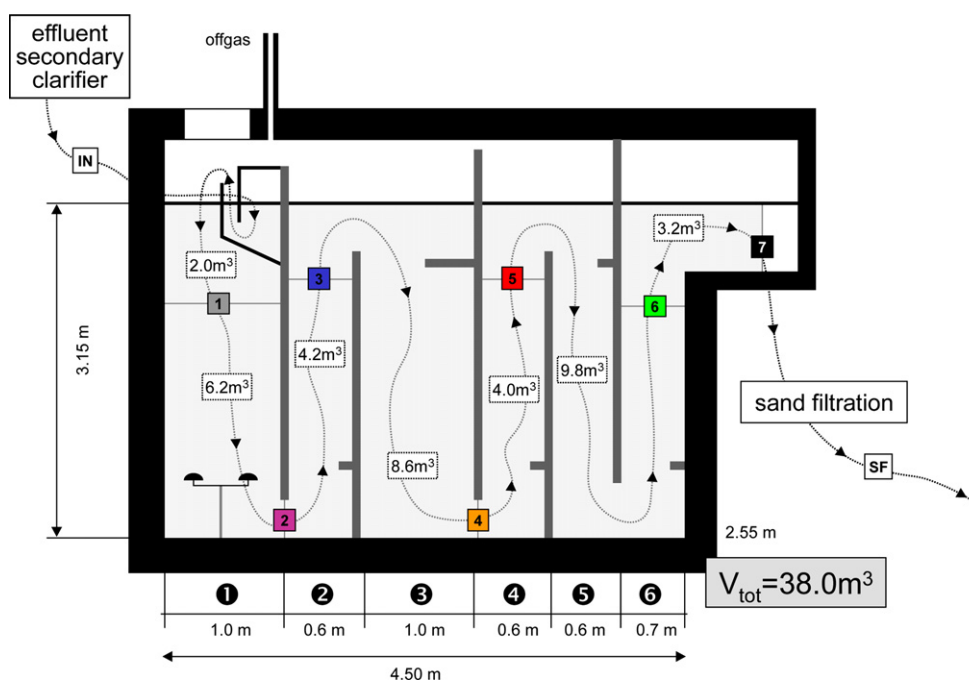


Fig. 1 – Scheme of the ozone reactor at WWTP Wüeri in Regensdorf, Switzerland.

diffusers into the first compartment. Since DOM – measured as dissolved organic carbon (DOC) – is the main matrix component determining the ozone consumption in wastewater, a DOC load-proportional ozone dosing was chosen to account for varying DOC concentrations in the secondary effluent. Hence, the flow to the ozone reactor was determined online by an ultrasonic device (Prosonic, Endress + Hauser, Germany), and the DOC concentration was measured online by an S::can sensor (S::can, Austria) using UV absorption. The S::can sensor was calibrated using grab samples filtered to $0.45\ \mu\text{m}$ and measured by a total organic carbon analyzer (Shimadzu, Switzerland). Seven sampling points (P1–P7) were distributed along the ozone reactor (Fig. 1), each consisting of a stainless steel pipe with an inner diameter of 5.5 mm reaching 80 cm into the ozone reactor. This resulted in residence times between 5 and 22 s in the sampling tubes, depending on the length and the flow within the tubes.

2.3. Tracer tests

The hydraulic behavior of the ozone reactor was characterized using fluoresceine as a conservative tracer during Dirac pulse tracer tests and as a reactive tracer during a continuous tracer test. Further information can be found in Text S1 of the SI.

2.4. Sampling campaigns

Seven ozone doses were applied ranging from 0.21 to $1.24\ \text{g O}_3\ \text{g}^{-1}\ \text{DOC}$ by adding a fixed gas volume of either 15 or $20\ \text{m}^3\ \text{h}^{-1}$ and by automatically adjusting the ozone concentration in the process gas (Table 1, Table S2 of the SI). Each ozone dose was injected for at least 20 h prior to sampling to guarantee equilibration of the sand filter (hydraulic retention time (HRT) 0.5 – 1 h). Grab samples were taken from the influent to the

ozone reactor (IN), along the ozone reactor (P2, P3, P7 or P3, P5, P7), and after sand filtration (SF). Depending on the ozone dose, either sampling point P2 or P5 was sampled. Dissolved ozone was quenched with sulfite (nitrosamine analysis) or nitrite (all other analyses) in at least 10-fold excess during sampling. Samples along the ozone reactor were taken accounting for the different travel times between sampling points in order to sample from the same water package each time.

2.5. Analyses of micropollutants and oxidation by-products

A total of 22 micropollutants (Table S1 of the SI without nitrosamines) were analyzed by online-SPE-LC-MS/MS (Hollender et al., 2009), while NDMA and NMOR (Table S1 of the SI) were extracted using offline SPE and analyzed by LC-MS/HRMS (Krauss and Hollender, 2008). Bromide and bromate concentrations were determined using ion chromatography followed by a post-column reaction (Salhi and von Gunten, 1999). Quenching the wastewater samples with sulfite (nitrosamines) or nitrite (micropollutants and bromide/bromate) did not influence quantification. Relative residual concentrations [%] were calculated by dividing the measured concentration at each sampling point with the concentration in the ozone reactor influent.

2.6. Disinfection and AOC measurement

The concentration of *Escherichia coli* (*E. coli*) was determined according to the Swiss standard method (Ettel, 2000). Total cell counts (TCC) were analyzed by flow cytometry using a nucleic acid staining by SYBR® Green (Hammes et al., 2008). The AOC concentration was measured by inoculation of the wastewater samples with a natural microbial consortium for 4 d at

Table 1 – Ozonation process and wastewater quality parameters of the influent grab samples collected during the sampling campaigns.

		Transferred ozone dose ^a [g O ₃ g ⁻¹ DOC]						
		0.21	0.41	0.60	0.74	0.81	0.90	1.24
Transferred ozone dose ^a	[mg O ₃ L ⁻¹]	0.8	1.2	2.6	3.0	3.8	4.3	5.2
Ozone exposure ^b	[M s]	n.d.	6.9 ± 0.7 × 10 ⁻⁴	3.1 ± 0.6 × 10 ⁻³	4.1 ± 0.4 × 10 ⁻³	6.1 ± 0.4 × 10 ⁻³	1.1 ± 0.1 × 10 ⁻²	1.9 ± 0.1 × 10 ⁻²
Ozone exposure ^b	[mg min L ⁻¹]	n.d.	0.5 ± 0.1	2.5 ± 0.5	3.3 ± 0.4	4.9 ± 0.4	8.9 ± 0.4	14.9 ± 0.7
k _{O₃}	[s ⁻¹]	n.d.	2.1 ± 0.3 × 10 ⁻²	9.3 ± 1.2 × 10 ⁻³	8.4 ± 0.7 × 10 ⁻³	6.0 ± 0.5 × 10 ⁻³	3.8 ± 0.0 × 10 ⁻³	3.6 ± 0.2 × 10 ⁻³
Rct	[-]	n.d.	1.3 ± 0.3 × 10 ⁻⁷	3.6 ± 0.2 × 10 ⁻⁸	4.6 ± 0.4 × 10 ⁻⁸	2.6 ± 0.1 × 10 ⁻⁸	1.7 ± 0.0 × 10 ⁻⁸	2.2 ± 0.3 × 10 ⁻⁸
Temperature	[°C]	20	20	13	20	13	13	22
DOC	[mg L ⁻¹]	3.4	2.4	4.4	3.4	4.7	4.8	4.1
pH ^c	[-]	7.0	7.0	7.0	7.0	7.0	7.0	7.2

n.d. – not determined.
a Applied ozone dose minus the ozone lost in the off-gas ((CO₃ process gas × Q_{process gas}) – (CO₃ offgas × Q_{offgas})/Q_{wastewater}/C_{DOC wastewater}).
b Measured average ozone exposure after complete ozone decay with standard deviation based on 2 or 3 laboratory batch experiments with the same wastewater.
c No change in pH by ozonation was observed.

30 °C and determination of the cell numbers by flow cytometry. The value of 1 × 10⁷ bacteria to 1 µg AOC L⁻¹ allowed the conversion of the natural microbial consortium growth into AOC concentrations (Hammes et al., 2006; Hammes and Egli, 2005).

2.7. Determination of pseudo first-order ozone decay constants, ozone and hydroxyl radical exposures and Rct values

The pseudo first-order ozone decay constants k_{O₃} as well as the ozone and hydroxyl radical exposures were determined by means of a laboratory dispenser system (Hoigné and Bader, 1994) (Table 1). *p*-Chlorobenzoic acid (*p*CBA) was spiked as ozone-resistant probe compound to measure the transient hydroxyl radical concentrations and to calculate the Rct values (Table 1). Further experimental details can be found in Text S2 of the SI.

2.8. Modeling of micropollutant oxidation and *E. coli* inactivation

The oxidation of a micropollutant P typically follows a second-order rate law (von Gunten, 2003b). During an ozonation process, both ozone and hydroxyl radicals have to be considered as oxidants:

$$-\frac{d[P]}{dt} = k_{P,O_3} [O_3] [P] + k_{P,\cdot OH} [\cdot OH] [P] \quad (1)$$

with k_{P,O₃} and k_{P,·OH} being second-order reaction rate constants for the reaction of the micropollutant with ozone and hydroxyl radicals, respectively, and [O₃] and [·OH] as ozone and hydroxyl radical concentrations.

To model micropollutant oxidation as well as *E. coli* inactivation in the full-scale ozone reactor, a four-step approach based on the combination of reactor hydraulics, laboratory-scale characterization of the wastewater with regard to ozone

chemistry, and reaction kinetics was used (von Gunten et al., 1997, 1999):

- Characterization of the hydraulics of the ozone reactor with a conservative tracer and fitting of the breakthrough curves by a series of ideal reactors (Section 3.1).
- Laboratory-scale experiments to determine the pseudo first-order rate constant for ozone decomposition and the Rct value (ratio between hydroxyl radical and ozone concentrations, (Elovitz and von Gunten, 1999)) at the respective wastewater pH and temperature (Section 2.7).
- Determination of the second-order rate constants for the oxidation of micropollutants by ozone (k_{P,O₃}) and hydroxyl radicals (k_{P,·OH}) and for *E. coli* inactivation by ozone for the respective wastewater pH and temperature. The second-order rate constants were compiled from literature (Table S3 of the SI) and corrected for pH by taking the acid-base speciation of the respective micropollutant into account and by using the species-specific reaction rate constants with ozone. Furthermore, the Arrhenius equation was used for temperature correction. An activation energy of 50 kJ mol⁻¹ was assumed which is in the range of typical activation energies for reactions with ozone (Hoigné and Bader, 1983).
- Coupling of reactor hydraulics with chemical kinetics by using software capable of solving coupled differential equations such as Berkeley Madonna (Macey et al., 2001).

The hydraulic model as well as eqs (1)–(3) were used to describe the chemical kinetics during ozonation and were solved simultaneously:

$$\text{Ozone decay: } -\frac{dO_3}{dt} = k_{O_3}[O_3] \quad (2)$$

$$\text{Hydroxyl radical concentration: } [\cdot OH] = Rct [O_3] \quad (3)$$

The arithmetic mean and standard deviation of the model were calculated by running Monte Carlo simulations (batch

runs with $n = 1000$) with k_{P,O_3} , $k_{P,\cdot OH}$, k_{O_3} and R_{ct} as random uniform distributions between the published or experimentally determined uncertainties of the respective parameter (Table 2).

2.9. Laboratory-scale batch experiments for micropollutant oxidation

In addition to the hydraulic model, a second approach based on laboratory-scale batch experiments was tested to predict micropollutant oxidation on the full-scale level. The concept included the use of wastewater from the same WWTP and the application of ozone under the same conditions (pH, T) as in the full-scale campaign, but with spiked micropollutants to allow for direct HPLC-UV/FL analysis. Accordingly, grab samples from the influent to the ozone reactor were taken, filtered (0.45 μm) and stored at 4 °C until use. Batch experiments were prepared and carried out as described in Section 2.7, however, with spiked micropollutants (0.2–1.0 μM) and by applying ozone doses yielding the same ozone exposures as in the full-scale campaigns. Relative residual concentrations of micropollutants were subsequently determined by HPLC-UV/FL, while ozone and hydroxyl radical exposures and R_{ct} values were determined as described in Section 2.7.

In addition, micropollutant transformation within the laboratory batch system was predicted according to eq (4), the integrated form of eq (1), which is only valid for a constant R_{ct} and for perfectly mixed and homogeneous solutions such as in a batch reactor:

$$\ln \frac{[P]}{[P_0]} = -k_{P,O_3} \int [O_3] dt - k_{P,\cdot OH} \int [\cdot OH] dt \\ = -(k_{P,O_3} + k_{P,\cdot OH} \cdot R_{ct}) \int [O_3] dt \quad (4)$$

with $\int [O_3] dt$ and $\int [\cdot OH] dt$ being the ozone and hydroxyl radical exposures, respectively. Eq (4) shows that the relative elimination of a micropollutant reacting with second-order kinetics during an ozonation process is concentration-independent with regard to micropollutants concentration, as long as the micropollutant does not significantly affect the oxidant stability.

3. Results and discussion

3.1. Hydraulics of the ozone reactor

The HRT of the ozone reactor varied between 3 and 15 min for storm (250 L s^{-1}) and dry weather conditions at night (35 L s^{-1}), respectively. For dry weather during daytime hours (80–100 L s^{-1}), the HRT was determined to be 7–8 min. The breakthrough curves derived from the Dirac pulse tracer tests could be modeled empirically for two flows as a series of five continuously stirred tank reactor (CSTR) cascades starting from P2 or P3 to P7 (Fig. 2). Table S4 of the SI summarizes the number of CSTRs in each cascade for the two flows and shows that the difference is marginal. Inclusion of a recirculation/back-mixing did not significantly improve the model fit. The hydraulic model started at P2 or P3, depending on the selected first sampling point within the ozone reactor for micropollutants or *E. coli* analysis. The first compartment of the ozone reactor was not included because short-circuiting occurred from P1 to P2 which was stronger when wastewater flows were $<150 \text{ L s}^{-1}$. In case of ideal mixing within a CSTR, breakthrough curves at P2 should decrease in height and increase in width as compared to P1. However, the breakthrough curve at P1 was only as high as and slightly wider than the breakthrough curve at P2 under these flow conditions (Fig. 2a). Results from a reactive continuous tracer test support the hypothesis of short-circuiting (Helbing et al., in preparation). In addition, $0.1 \pm 0.02\%$ of the tracer mass was detected prior to the main breakthrough curves at P7 (Figure S1 of the SI), but not at P1–P6. This hints on short-circuiting of a water fraction through the ozone reactor directly to P7, most likely on top of the reactor along the concrete walls. It remains unclear whether this small water fraction was exposed to ozone or not.

3.2. Kinetics of full-scale micropollutant oxidation

Fig. 3 shows the relative residual concentrations of three representative micropollutants with different second-order rate constants for their reaction with ozone. The fast-

Table 2 – Ozone chemistry parameters used within the Monte Carlo simulations for ozone doses of 0.41 and 0.74 $\text{g O}_3 \text{ g}^{-1}$ DOC, including the respective range of their random uniform distributions.

		Transferred ozone dose ^a [$\text{g O}_3 \text{ g}^{-1}$ DOC]		Reference
		0.41	0.74	
$k_{O_3, \text{benzotriazole}}$ ^b	$[\text{M}^{-1} \text{ s}^{-1}]$	$1.86\text{--}2.70 \times 10^2$		Lutze (2005)
$k_{\cdot OH, \text{benzotriazole}}$ ^c	$[\text{M}^{-1} \text{ s}^{-1}]$	$7.10\text{--}8.10 \times 10^9$		Naik and Moorthy (1995)
$k_{O_3, \text{atenolol}}$	$[\text{M}^{-1} \text{ s}^{-1}]$	$1.30\text{--}2.10 \times 10^3$		Benner et al. (2008)
$k_{\cdot OH, \text{atenolol}}$	$[\text{M}^{-1} \text{ s}^{-1}]$	$7.50\text{--}8.50 \times 10^9$		Benner et al. (2008)
k_{O_3}	$[\text{s}^{-1}]$	$1.97\text{--}2.42 \times 10^{-2}$	$7.60\text{--}8.80 \times 10^{-3}$	Present study
R_{ct}	$[-]$	$1.14\text{--}4.71 \times 10^{-7}$	$4.26\text{--}28.50 \times 10^{-8}$	Present study

a Applied ozone dose minus the ozone lost in the off-gas ($(\text{CO}_2 \text{ process gas} \times Q_{\text{process gas}}) - (\text{CO}_2 \text{ offgas} \times Q_{\text{offgas}}) / Q_{\text{wastewater}} / \text{CDOC wastewater}$).

b $k_{O_3, \text{benzotriazole}}$ determined at pH 2 and 5 (18 and $22 \text{ M}^{-1} \text{ s}^{-1}$, respectively, Karpel Vel Leitner and Roshani, 2010) support the $k_{O_3, \text{benzotriazole}}$ value determined by Lutze (2005) at pH 7. Standard deviation of $k_{O_3, \text{benzotriazole}}$ at pH 7 was set as difference between the measured ($2.28 \times 10^2 \text{ M}^{-1} \text{ s}^{-1}$) and modeled ($1.86 \times 10^2 \text{ M}^{-1} \text{ s}^{-1}$) value determined in Lutze (2005), and the measured value was used as average value.

c Standard deviation of $k_{\cdot OH, \text{benzotriazole}}$ was estimated to be equal to the standard deviation of $k_{\cdot OH, \text{atenolol}}$.

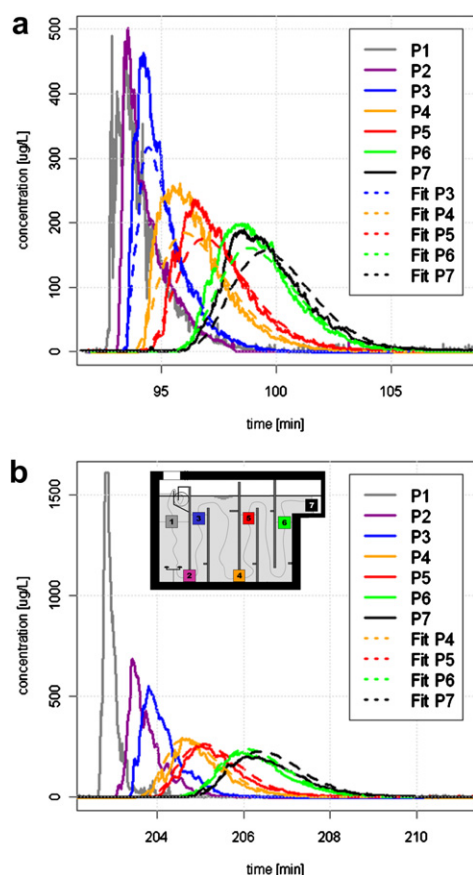


Fig. 2 – Breakthrough curves from the Dirac pulse tracer test of the conservative tracer fluoresceine at the ozone reactor for wastewater flows of a) 87 L s^{-1} and b) 170 L s^{-1} , both with $15 \text{ m}^3 \text{ h}^{-1}$ oxygen gas flow. Experimental data are shown as solid curves for P1–P7 and hydraulic model calculations are depicted as dashed lines from a) P3 and b) P4 to P7.

reacting substance diclofenac with a $k_{\text{O}_3} > 10^4 \text{ M}^{-1} \text{ s}^{-1}$ (Table S3 of the SI) was completely oxidized for ozone doses $\geq 0.41 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$ within the first compartment of the ozone reactor (Fig. 3a). Based on eq (4), this was expected from the ozone exposures (Table 1) and its k_{O_3} . Surprisingly, $\sim 8\%$ of diclofenac resisted ozonation at a dose of $0.21 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$. According to eq (4), an ozone exposure of $6.7 \times 10^{-6} \text{ M s}$ is needed to achieve 99% oxidation of diclofenac with $k_{\text{O}_3} = 6.8 \times 10^5 \text{ M}^{-1} \text{ s}^{-1}$. Although the ozone exposure resulting from the $0.21 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$ dose could not be determined by means of the laboratory dispenser system (Hoigné and Bader, 1994), because ozone was consumed too fast, the ozone exposure needed for 99% oxidation is less than 1% of the final exposure of the $0.41 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$ dose. Hence, a 99% oxidation was expected even for an ozone dose of $0.21 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$. The incomplete oxidation of diclofenac and other fast-reacting substances such as carbamazepine, sulfamethoxazole and clarithromycin (Figure S2 of the SI), can be attributed to short-circuiting from the first into the second compartment (Section 3.1). 8–21% of the mentioned micropollutants were not oxidized at a dose of $0.21 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$ which is in

agreement with the percentage of non-oxidized water (about 15%) obtained during the reactive continuous fluoresceine tracer test carried out under identical conditions (Helbing et al., in preparation). The short-circuiting is a hydraulic phenomenon and therefore independent of the ozone dose. However, for the $0.21 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$ dose, dissolved ozone was completely consumed up to P2, leading to the observed residual fraction of fast-reacting micropollutants due to non-ozonated water packages. For all higher ozone doses, residual ozone was measured at P2. This may have led to a mixing of water packages containing dissolved ozone with others not containing ozone before reaching P2. As a consequence, the intrinsically fast oxidation of the fast-reacting micropollutants resulted in their complete oxidation.

The other fast-reacting substances clindamycin, erythromycin, mefenamic acid, naproxen, sulfapyridine and trimethoprim as well as sotalol ($k_{\text{O}_3} > 10^3 \text{ M}^{-1} \text{ s}^{-1}$), were oxidized below their limit of quantification (LOQ) (Table S1 of the SI) at a dose of $0.21 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$ at P2. This is not caused by a different oxidation behavior as compared to diclofenac, carbamazepine or sulfamethoxazole, but by their low influent concentrations which render them non-detectable even for $\leq 70\%$ relative residual concentration. Mefenamic acid with an unknown k_{O_3} was classified as fast-reacting compound from its structure. Its k_{O_3} was estimated to be higher than that of diclofenac because the two methyl groups increase the electron density of the aromatic system in contrast to the two electron-withdrawing chlorine atoms of diclofenac (Table S1 of the SI).

Micropollutants reacting slower with ozone ($k_{\text{P},\text{O}_3} < 10^4 \text{ M}^{-1} \text{ s}^{-1}$), such as atenolol and benzotriazole, were oxidized along the ozone reactor with increasing ozone and hydroxyl radical exposure (Fig. 3b and c). The degree of transformation also increased with increasing k_{O_3} ($k_{\text{O}_3, \text{atenolol}} = 1.7 \times 10^3 \text{ M}^{-1} \text{ s}^{-1} > k_{\text{O}_3, \text{benzotriazole}} = 2.3 \times 10^2 \text{ M}^{-1} \text{ s}^{-1}$ at pH 7). Ibuprofen, paracetamol and sulfamethazine were not detected in the influent of the ozone reactor, and sulfadiazine and phenazone were only detected once and as expected fully oxidized at a dose of 0.81 and $1.24 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$, respectively. The slowly-reacting bezafibrate was detected in the influent on three sampling days, and was oxidized below the LOQ at the three highest ozone doses $\geq 0.60 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$ (Figure S2 of the SI).

3.3. Prediction of micropollutant oxidation based on a model coupling reactor hydraulics with ozone chemistry and reaction kinetics

The modeled ozone and hydroxyl radical concentration profiles for the two ozone doses 0.41 and $0.74 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$ are illustrated in Fig. 4. The relatively high number of reactors in the CSTR cascades (Table S4 of the SI) indicates that the ozone reactor behaves similar to a plug flow reactor (PFR) after the first compartment. Hence, the ozone reactor was also modeled as a PFR by combining eqs (1)–(3) to determine the influence of the hydraulic model on the prediction of ozone concentration profiles and micropollutant oxidation. The CSTR and PFR models predicted very similar ozone profiles, showing that the hydraulic model structure was of minor importance for this specific ozone reactor. Resulting ozone exposures were predicted to be 60% ($0.41 \text{ g O}_3 \text{ g}^{-1}$

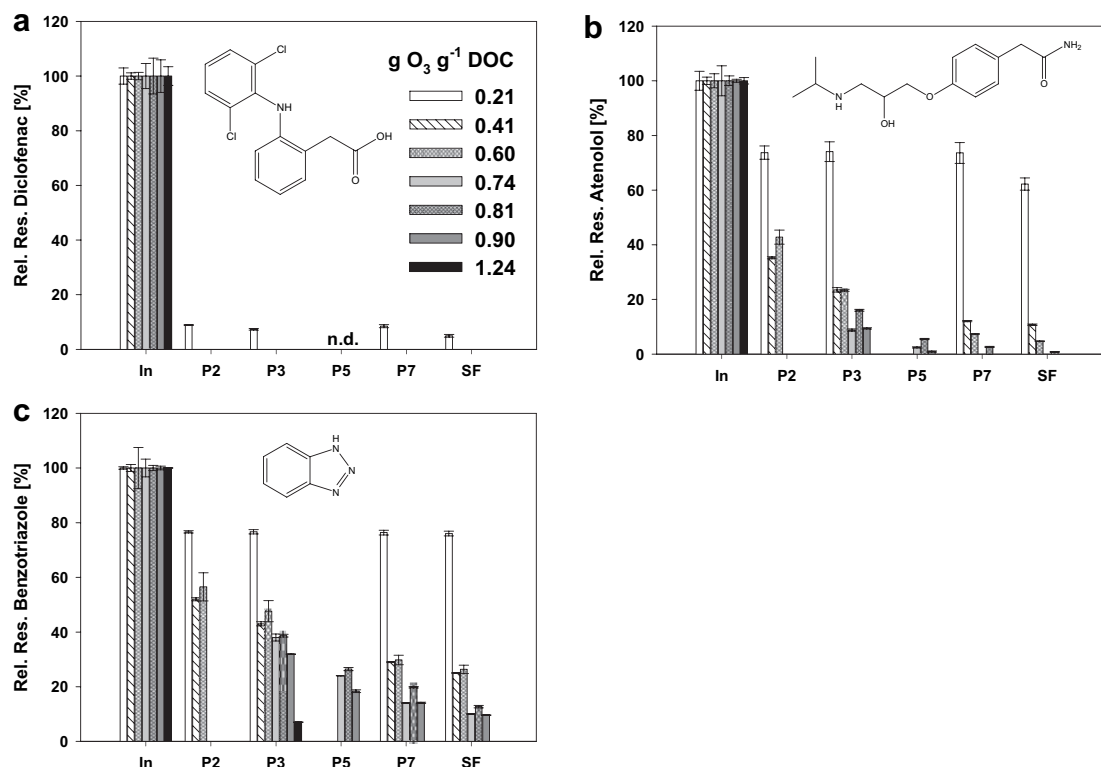


Fig. 3 – Relative residual concentrations of a) diclofenac, b) atenolol and c) benzotriazole for seven ozone doses ranging from 0.21 to 1.24 g O₃ g⁻¹ DOC in the influent to the ozone reactor (IN), along the ozone reactor (P2, P3, P7 or P3, P5, P7) and after sand filtration (SF). Error bars were calculated by linear error propagation derived from duplicate measurements. n.d. = not determined.

DOC) and 26% (0.74 g O₃ g⁻¹ DOC) lower by the hydraulic models than for the full-scale measurements (Fig. 4). Based on the lower ozone exposures and eq. (3), the modeled hydroxyl radical exposures were significantly smaller than the effective exposures in the full-scale system. The different ozone addition and mixing in the full-scale reactor (bubble column, non-homogeneously mixed) compared to the batch reactor (dosage of a stock solution, homogeneously mixed) may have affected the ozone behavior during the second phase of ozone decay, leading to different k_{O_3} values in both set-ups. However, differences in the initial phase of ozone decay (first ~20 s, Buffle et al., 2006) can be excluded as a factor for the observed deviations of the ozone profiles because the ozone decay in the full-scale ozone reactor already followed the second phase of ozone decay at the starting point of the hydraulic model (P2: HRT 18 s relative to P1), and can hence be described by k_{O_3} (von Gunten, 2003b).

Fig. 5 shows the results of the CSTR and PFR models for benzotriazole and atenolol. As for the ozone profiles, predictions by the PFR model were very similar to the CSTR model. For an ozone dose of 0.41 g O₃ g⁻¹ DOC, the measured and predicted values correlated well, while for an ozone dose of 0.74 g O₃ g⁻¹ DOC, model predictions were about a factor of 2.5 higher than full-scale measurements. Monte Carlo simulations showed that deviations between measurements and model predictions were larger than model uncertainties (standard deviations CSTR and PFR models Fig. 5). This is an

independent evidence that the model overestimated micropollutant oxidation for the ozone dose of 0.74 g O₃ g⁻¹ DOC.

The underestimation of oxidant exposures (Fig. 4) should have led to underestimation of micropollutant oxidation in the model predictions. However, the contrary was observed. Therefore, counteracting processes must have lowered micropollutant oxidation in wastewater at the full-scale level, to compensate for the underestimation of oxidant exposures in the model predictions. Huber et al. (2005) attributed the overestimation of micropollutant oxidation based on ozone and hydroxyl radical exposures to sorption of some compounds to sludge particles or the interaction with colloids which might offer some protection against ozone attack. We did not distinguish between micropollutants in the colloidal (particles between 1 nm and 1 µm) and dissolved phase in this study, but micropollutants in both phases were concentrated on and eluted from SPE cartridges (Zhang and Zhou, 2007). The deviations between model predictions and measurements for benzotriazole and atenolol at a dose of 0.74 g O₃ g⁻¹ DOC correspond to 45% and 19% at P5, respectively, and to 22% for benzotriazole at P7. These deviations are in the range of the percentage of chemicals sorbed to aquatic colloids (Maskaoui et al., 2007; Zhou et al., 2007). Since both benzotriazole and atenolol show sorption capabilities (Hinterbuchner, 2006; Yamamoto et al., 2009), it might be possible that these micropollutants were protected from ozone attack by the interaction with aquatic colloids.

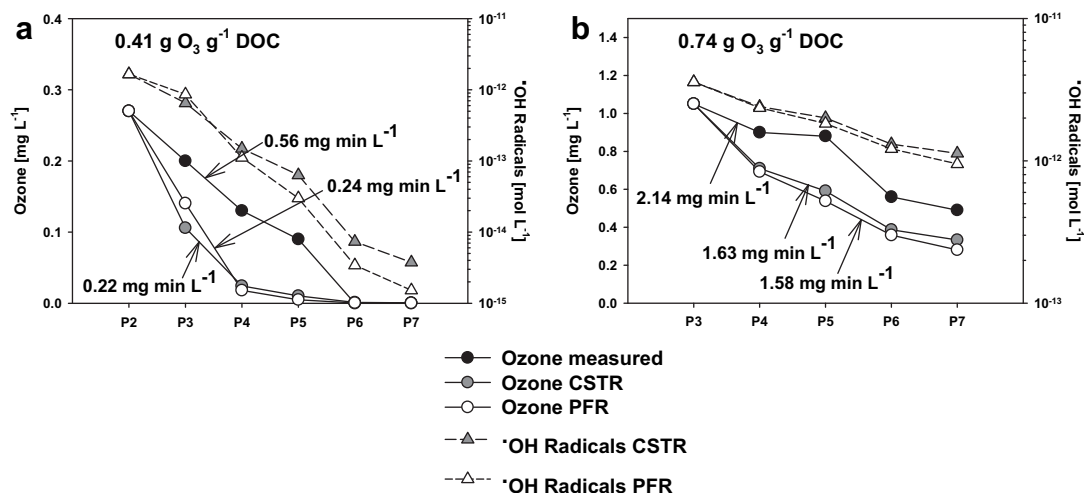


Fig. 4 – Measured ozone (black circles) and modeled ozone and hydroxyl radical profiles for the continuously stirred tank reactor (CSTR) model (grey circles and triangles) and the plug flow reactor (PFR) model (open circles and triangles) along the ozone reactor (P2/P3–P7) for an ozone dose of a) $0.41 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$ and b) $0.74 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$. Measured and modeled standard deviations are given but smaller than the respective symbols. The ozone exposures from P2/P3 to P7 are indicated in mg min L^{-1} . Storm weather conditions occurred during the $0.74 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$ campaign, leading to a HRT of only 3–4 min (Table S2) and a DOC of 3.4 mg L^{-1} (Table 1).

In contrast to these slower-reacting substances, fast-reacting substances with a $k_{\text{O}_3} > 10^4 \text{ M}^{-1} \text{ s}^{-1}$ were already completely oxidized within the first compartment of the ozone reactor (Section 3.2). This may be attributed to their intrinsic fast oxidation, leading to lower concentrations in the aqueous phase. This in turn causes a shift in the colloid/micropollutant equilibrium leading to quick desorption of micropollutants from colloids. Maskaoui et al. (2007) determined 5 min to be sufficient to reach the colloid/pharmaceutical equilibrium. Although desorption kinetics have not been reported, quick desorption and subsequent fast oxidation by ozone are possible, leading to the observed oxidation of fast-reacting micropollutants below the LOQ in both the colloidal and dissolved phase. The same processes (sorption equilibrium, oxidation, desorption, further oxidation) apply to slower-reacting micropollutants as well. However, micropollutant desorption from the colloidal to the aqueous phase is much slower due to a slower oxidation of these compounds in aqueous solution.

In addition, model predictions were calculated based on minimum and maximum Rct values (1×10^{-8} – 5×10^{-7}) determined in secondary effluent from WWTP Wüeri sampled under varying weather conditions over a period of 16 months (light grey bars, Fig. 5). These calculations show that the Rct value is highly important for the oxidation of slowly-reacting micropollutants, since hydroxyl radicals contribute significantly to their oxidation. Predictions for the relative residual concentrations at P7 for the two modeled ozone doses varied from 39 to 93% and 0 to 58% for benzotriazole, respectively, and from 27 to 64% and 0 to 6% for atenolol, respectively. Hence, differences between the Rct value in the full-scale reactor and laboratory batch experiments could directly contribute to the observed variations. Different Rct values may be derived from different

modes of ozone addition and mixing in the full-scale reactor (bubble column, non-homogeneously mixed) compared to the batch experiment (dosage of a stock solution, homogeneously mixed).

In conclusion, the two counteracting processes i) underestimation of modeled oxidant exposures (leading to an underestimation of micropollutants oxidation in the hydraulic model) and ii) protection of the sorbed micropollutants fraction from ozone attack (leading to an overestimation of micropollutants oxidation in the hydraulic model) may have compensated each other to varying extents. In combination with possibly different Rct values in the full-scale vs. batch reactor, this resulted in a good prediction of micropollutants oxidation for the $0.41 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$ dose, and in an overestimation within the factor of 2.5 for the $0.74 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$ dose. The factor of 2.5 hence represents the range of possible deviations of the hydraulic model from the full-scale measurements at the ozone reactor at WWTP Wüeri in Regensdorf.

3.4. Prediction of full-scale micropollutant oxidation by laboratory-scale batch experiments

Fig. 5 illustrates that micropollutant oxidation derived from laboratory-scale batch experiments and the respective laboratory-scale predictions based on eq. (4) are in good agreement, but deviate up to a factor of 1.5 from full-scale measurements. Since different wastewater grab samples were used during laboratory- and full-scale experiments, and the comparison was based on the same ozone exposures, the Rct value was identified as a major source for the observed deviations. It was 3–5 times lower in the batch experiments (data not shown) compared to batch experiments with wastewater from the full-scale sampling campaigns. Hence, it is highly

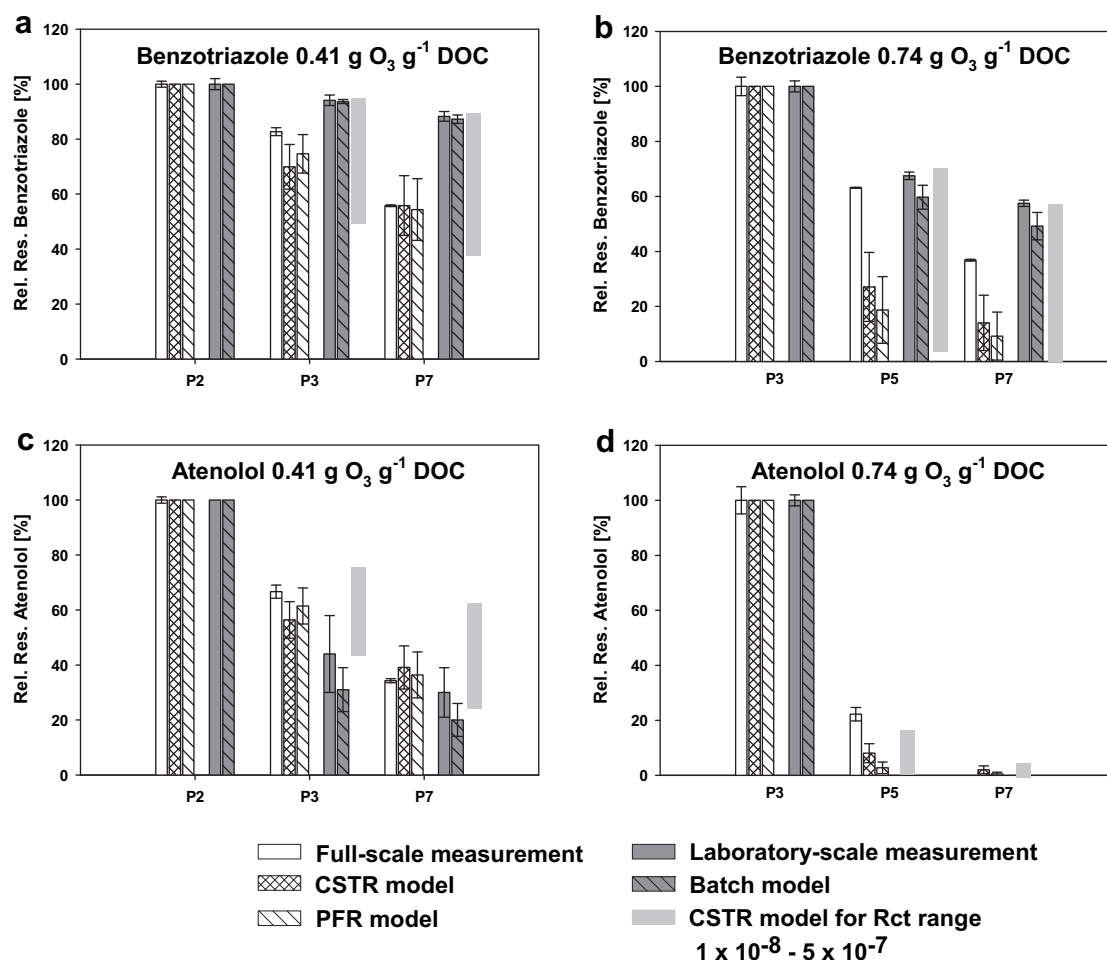


Fig. 5 – Relative residual micropollutant concentrations in various scenarios for benzotriazole with a) 0.41 and b) 0.74 g O₃ g⁻¹ DOC and for atenolol with c) 0.41 and d) 0.74 g O₃ g⁻¹ DOC starting from sampling point P2 or P3. Abbreviations: CSTR Model = Continuously Stirred Tank Reactor Model; PFR Model = Plug Flow Reactor Model. Laboratory-scale measurements were carried out in a batch experiment with secondary effluent from WWTP Wüeri under the same experimental conditions (ozone dose, pH, temperature) as in the full-scale measurement. The batch model is based on oxidant exposures determined during laboratory-scale measurements. Error bars from full- and laboratory-scale measurements were calculated by linear error propagation derived from duplicate measurements. Standard deviations of CSTR and PFR model predictions were calculated by Monte Carlo simulations.

important to determine the possible range of Rct values, which strongly depends on the organic and inorganic matrix, of a specific wastewater under all conditions (rain, temperature, season, daytime). Due to the important contribution of hydroxyl radicals to the oxidation of slowly-reacting micropollutants, the Rct is a critical parameter for these cases. This is underlined by a recent study investigating the uncertainty during micropollutant ozonation (Neumann et al., 2009). The Rct value (and not k_{P,O_3} or $k_{P,OH}$) was found to be the most influential parameter for slowly-reacting substances, while the accuracy of the hydraulic model of a reactor was determined to be most important for fast-reacting substances.

Finally, the complete oxidation of the fast-reacting substances carbamazepine, diclofenac, sulfamethoxazole and 17 α -ethinylestradiol ($k_{O_3,17\alpha\text{-ethinylestradiol}} = 1.6 \times 10^6 \text{ M}^{-1} \text{ s}^{-1}$ at pH 7 (Deborde et al., 2005)) in the laboratory-scale batch experiments was achieved for ozone exposures occurring at

P2/P3 during full-scale sampling campaigns (data not shown), and is in accordance with full-scale measurements. Micropollutants interacting with colloids were of minor importance within the batch experiments since their respective concentrations in the aqueous phase were much higher (spiking of micropollutants), and accordingly, the fraction of micropollutants interacting with colloids was much lower. In addition, low residual concentrations could not be detected due to higher LOQs of the LC-UV/FL method as compared to the online-SPE-LC-MS/MS method.

3.5. Oxidation by-products

Bromate, a potential human carcinogen, forms during ozonation of bromide-containing waters through a combination of reactions involving ozone and secondary oxidants such as hydroxyl and carbonate radicals (von Gunten, 2003a). Bromide

levels in the influent to the ozone reactor varied from $6 \mu\text{g L}^{-1}$ (due to dilution) to $32 \mu\text{g L}^{-1}$ (Table S2 of the SI). These values are in accordance with low bromide levels ($<5\text{--}50 \mu\text{g L}^{-1}$) found in a survey of 85 water resources for drinking water treatment in Switzerland (von Gunten and Salhi, 2003). Fig. 6a shows that only ozone doses $\geq 0.74 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$ yielded bromate concentrations above the LOQ of $2 \mu\text{g L}^{-1}$. Bromate formation is a slow process (von Gunten, 2003a), reflected in a bromate increase from P3 to P7 for the highest ozone dose. As expected, aerobic sand filtration did not remove bromate. The final bromate concentrations (max. $7.5 \mu\text{g L}^{-1}$) were below the drinking water standard of $10 \mu\text{g L}^{-1}$ for all ozone doses and hence also below a proposed ecotoxicological threshold value of 3 mg L^{-1} (Hutchinson et al., 1997).

The concentrations of NDMA and NMOR in the reactor are illustrated in Fig. 6b for the highest ozone dose ($1.24 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$). A previous study showed that NDMA formation from secondary effluent did not correlate with the ozone dose (Hollender et al., 2009) and might strongly depend on the concentrations and conversion rate of precursors as well as matrix composition. In the present study, NDMA formation was observed to be a fast process (strong increase up to P3), which was followed by a slight decrease within the ozone reactor to P7. The fast formation of NDMA points towards specific precursors rather than moieties of DOM as precursors (von Gunten et al., 2010). As observed in a previous study (Hollender et al., 2009), sand filtration resulted in a decrease of NDMA concentration due to biodegradation, leading to concentrations slightly above the notification level of 10 ng L^{-1} for NDMA in drinking water in California (CDPH, 2009). An ecotoxicological threshold value does currently not exist for NDMA. In contrast to NDMA, neither ozonation nor sand filtration did have an influence on NMOR concentrations (Fig. 6b).

3.6. AOC formation

Fig. 7a shows that the AOC levels in the secondary effluent varied between 40 and $200 \mu\text{g L}^{-1}$, which is in the range of

concentrations found in lake water (Hammes et al., 2006). Hence, AOC was either not completely consumed during biological wastewater treatment or it might have been released from lysating cells due to a natural turnover of the microbial sewage sludge community. During ozonation, AOC levels increased up to a factor of 6, reaching a maximum concentration of $740 \mu\text{g L}^{-1}$. AOC formation was a fast process since no substantial increase was observed after the initial increase at P1. Similar results were found during ozonation of lake water (Hammes et al., 2006). Still, no clear trend could be found when correlating AOC formation with the ozone dose, ozone exposure or influent AOC concentrations. The rapid sand filter finally decreased the AOC levels by 14–46%, leading to AOC concentrations above the level before ozonation in some cases, as previously observed for ozonated drinking water followed by GAC/sand filtration (van der Kooij et al., 1989). Interestingly, the AOC levels after sand filtration depended on the wastewater temperature, and not on the AOC levels prior to sand filtration. AOC levels after sand filtration of $390\text{--}440 \mu\text{g L}^{-1}$ were found in April 2008 (wastewater temperature 13°C), while $120\text{--}150 \mu\text{g L}^{-1}$ AOC were determined in August 2008 (wastewater temperature $20\text{--}22^\circ\text{C}$).

3.7. Removal of TCC and *E. coli*

Ozone is an excellent disinfectant and inactivates a broad range of microorganisms (von Gunten, 2003a). TCC in the influent to the ozone reactor were relatively stable and varied from 3.3 to $8.4 \times 10^6 \text{ cells mL}^{-1}$ (Fig. 7b), as determined by flow cytometry. Ozonation decreased TCC by 1–4.5 log units, not consistently correlating with the ozone doses. A distinct regrowth of TCC was observed during sand filtration, since AOC formed during ozonation could act as a carbon source for microorganisms which were subsequently released from the sand filter. The release of TCC from the sand filter seemed to be independent of the TCC or AOC level after ozonation (Fig. 7) and ΔTCC could not be correlated with ΔAOC concentrations (data not shown). Overall, TCC could be

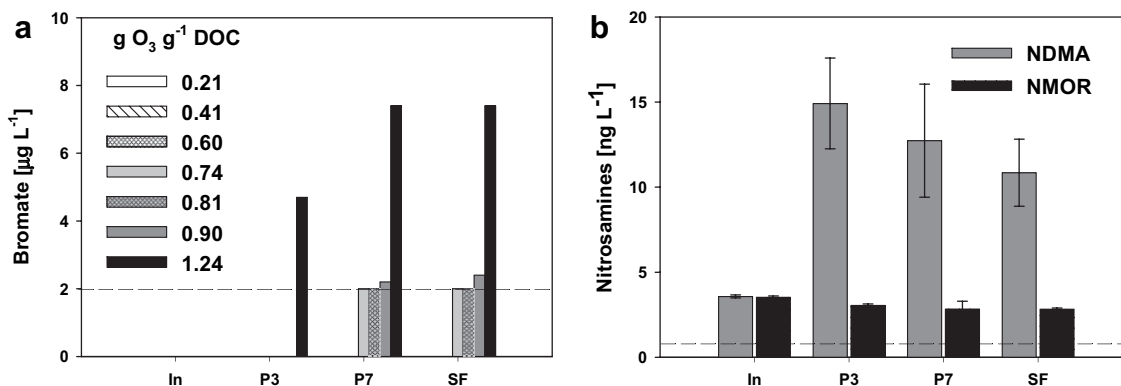


Fig. 6 – Oxidation by-products concentrations in the influent to the ozone reactor (IN), along the ozone reactor (P3, P7) and after sand filtration (SF): a) Bromate concentrations for seven ozone doses ranging from 0.21 to $1.24 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$. The limit of quantification (LOQ) was $2 \mu\text{g L}^{-1}$ and bromide concentrations varied between 6 and $32 \mu\text{g L}^{-1}$ (Table S2 of the SI). b) N-Nitrosodimethylamine (NDMA) and N-Nitrosomorpholine (NMOR) concentrations for an ozone dose of $1.24 \text{ g O}_3 \text{ g}^{-1} \text{ DOC}$ (LOQ = 1 ng L^{-1}). Error bars were calculated by linear error propagation derived from duplicate measurements. Note the different concentration units in Figures a) and b).

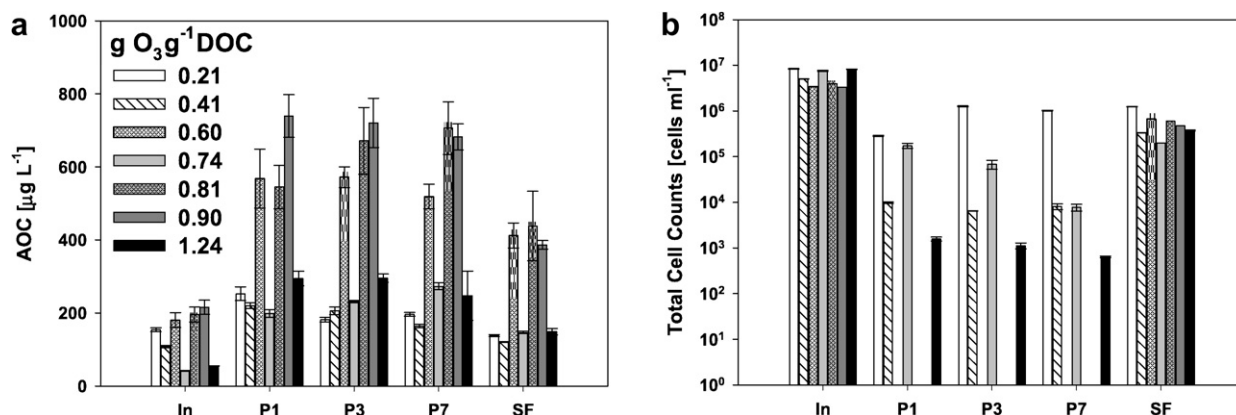


Fig. 7 – a) Assimilable organic carbon (AOC) concentrations and b) total cell counts for seven ozone doses ranging from 0.21 to 1.24 $\text{g O}_3 \text{ g}^{-1} \text{DOC}$ in the influent to the ozone reactor (IN), along the ozone reactor (P1, P3, P7) and after sand filtration (SF). Error bars indicate standard deviations based on triplicate measurements.

lowered by 0.5–1.5 log units by the combination of ozonation and sand filtration.

E. coli concentrations were lowered by 0.5–2.5 log units during ozonation (Fig. 8, Table S2 of the SI), and only partially correlated with the ozone exposure (Table 1). After a fast initial inactivation during the first compartment of the ozone reactor up to P3, *E. coli* was not further eliminated substantially from P3 to P7. This was unexpected for ozone doses $\geq 0.41 \text{ g O}_3 \text{ g}^{-1} \text{DOC}$ since dissolved ozone was still apparent from P3 to P7. *E. coli* inactivation by ozone is a fast process and well predicted in drinking water treatment based on the concept of reactor hydraulics coupled with ozone chemistry and reaction kinetics (von Gunten, 2003a). However, in the present study, the same concept applied to wastewater treatment strongly overestimated *E. coli* inactivation (Figure S3 of the SI). The water fraction of 0.1% short-circuiting directly to P7 cannot be the main reason for the large discrepancy. In a worst case scenario with 0.1% of the water not getting in contact with ozone at all, a maximum *E. coli* inactivation of 3 log units would be achieved. In case of the

short-circuiting water packages getting in contact with ozone, the overall expected inactivation should be higher. However, a maximum inactivation of ~ 0.5 log units from P3 to P7 was observed (Figure S3 of the SI). The large differences between measured and modeled *E. coli* inactivation could be explained by the shielding of *E. coli* in activated sludge flocs, which can experience lower ozone exposures than the aqueous solution (Huber et al., 2005). Accordingly, *E. coli* present in activated sludge flocs might not be inactivated during ozonation, but could be released from the activated sludge flocs at a later stage and lead to a positive result in plating techniques. Finally, no regrowth of *E. coli* was observed upon sand filtration. Combining ozonation and sand filtration is hence an efficient barrier against *E. coli* and possibly other pathogens.

4. Conclusions

- The predicted oxidation of slowly-reacting micropollutants based on coupling reactor hydraulics with ozone chemistry and reaction kinetics was up to a factor of 2.5 higher than full-scale measurements. This overestimation was attributed to a protection of micropollutants from ozone attack by the interaction with aquatic colloids in the real system.
- Laboratory-scale batch experiments using wastewater from the same full-scale WWTP could predict the oxidation of slowly-reacting micropollutants ($k_{\text{O}_3} < 10^4 \text{ M}^{-1} \text{ s}^{-1}$) on the full-scale level within a factor of 1.5. The R_{ct} value (depending strongly on the organic and inorganic wastewater matrix) was identified as a major source to this deviation. Hence, it is important to determine the spectrum of R_{ct} values for various conditions in wastewater of a specific WWTP. Oxidation of fast-reacting substances could be well predicted since their oxidation does not significantly depend on the formation of hydroxyl radicals.
- An increase in bromate and NDMA concentrations to non-problematic levels was observed during ozonation, while no influence on NMOR concentrations could be shown. AOC was formed during ozonation of wastewater of up to $740 \mu\text{g L}^{-1}$ and subsequently degraded by up to 46% during

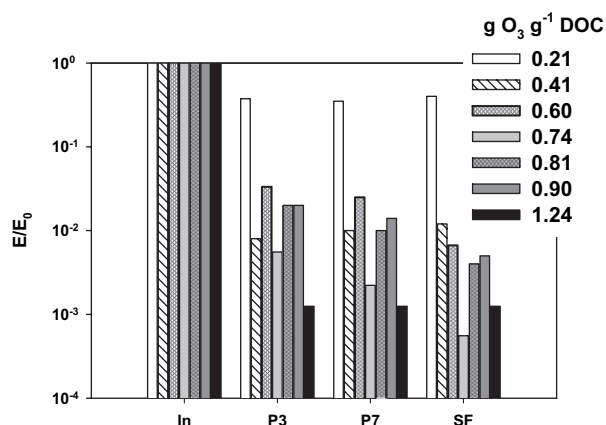


Fig. 8 – Relative residual concentrations of *E. coli* for seven ozone doses ranging from 0.21 to 1.24 $\text{g O}_3 \text{ g}^{-1} \text{DOC}$ in the influent to the ozone reactor (IN), along the ozone reactor (P3–P7) and after sand filtration (SF).

rapid sand filtration. Along with the slight decrease of NDMA during sand filtration, this finding highlights the importance of sand filtration after an ozonation step for removal of oxidation by-products (AOC, NDMA).

- Disinfection during ozonation in the full-scale reactor was demonstrated by the fast decrease of the TCC and inactivation of *E. coli*. However, predictions of *E. coli* inactivation did strongly deviate from full-scale measurements, which can be attributed to the shielding of *E. coli* by activated sludge flocs.

Acknowledgements

This study was part of the project Strategy MicroPoll funded by the Swiss Federal Office for the Environment (FOEN). Support for S.G. Zimmermann was provided by the EU project Neptune (Contract No 036845, SUSTDEV-2005-3.II.3.2), which was financially supported by grants obtained from the EU Commission within the Energy, Global Change and Ecosystems Program of the Sixth Framework (FP6-2005-Global-4). The authors gratefully acknowledge S. Brocker and D. Rensch for installation and help with maintenance of the ozone reactor, the staff at WWTP Wüeri for general support, S. Koepke for SPE-LC-MS/MS measurements, E. Salhi for laboratory assistance and bromate measurements, F. Hammes and H. P. Fuchsli for AOC and TCC determination, E. Gansner and M. Koch for *E. coli* measurements, J. Helbing and C. Abegglen for help with the tracer tests, M. Gresch for support during modeling, and J.L. Kormos for language corrections. The authors also thank CWQRC at Curtin University, Australia, and BfG, Germany, where the modeling and writing took place, Jaehong Kim for fruitful discussions as well as two anonymous reviewers for their helpful comments.

Appendix. Supplementary material

Supplementary data associated with this article can be found in the online version, at doi:10.1016/j.watres.2010.07.080.

REFERENCES

- Benner, J., Salhi, E., Ternes, T., von Gunten, U., 2008. Ozonation of reverse osmosis concentrate: kinetics and efficiency of beta blocker oxidation. *Water Res.* 42 (12), 3003–3012.
- Boller, M., von Gunten, U., Pianta, R., Solcà, L., 2000. Modelling full-scale advanced micropollutant oxidation. In: 9th Gothenburg Symposium: Chemical Water and Wastewater Treatment VI, Istanbul, Turkey.
- Buffle, M.O., Schumacher, J., Salhi, E., Jekel, M., von Gunten, U., 2006. Measurement of the initial phase of ozone decomposition in water and wastewater by means of a continuous quench-flow system: application to disinfection and pharmaceutical oxidation. *Water Res.* 40 (9), 1884–1894.
- CDPH, 2009. NDMA and Other Nitrosamines – Drinking Water Issues. California Department of Public Health. <http://www.cdph.ca.gov/certlic/drinkingwater/Pages/NDMA.aspx> December 15.
- Deborde, M., Rabouan, S., Duguet, J.-P., Legube, B., 2005. Kinetics of aqueous ozone-induced oxidation of some endocrine disruptors. *Environ. Sci. Technol.* 39 (16), 6086–6092.
- Dodd, M.C., Kohler, H.P.E., von Gunten, U., 2009. Oxidation of antibacterial compounds by ozone and hydroxyl radical: elimination of biological activity during aqueous ozonation processes. *Environ. Sci. Technol.* 43 (7), 2498–2504.
- Elovitz, M.S., von Gunten, U., 1999. Hydroxyl radical ozone ratios during ozonation processes. I – the R-ct concept. *Ozone-Sci. Eng.* 21 (3), 239–260.
- Escher, B.I., Bramaz, N., Ort, C., 2009. Monitoring the treatment efficiency of a full scale ozonation on a sewage treatment plant with a mode-of-action based test battery. *J. Environ. Monit.* 11 (10), 1836–1846.
- Mikrobiologie. In: Ettel, W. (Ed.), Schweizerisches Lebensmittelhandbuch. Bundesamt für Gesundheit, Bern (Kapitel 56).
- Gallard, H., von Gunten, U., Kaiser, H.P., 2003. Prediction of the disinfection and oxidation efficiency of full-scale ozone reactors. *J. Water Supply Res. Technol. AQUA* 52 (4), 277–290.
- von Gunten, U., 2003a. Ozonation of drinking water: part II. Disinfection and by-product formation in presence of bromide, iodide or chlorine. *Water Res.* 37 (7), 1469–1487.
- von Gunten, U., 2003b. Ozonation of drinking water: part I. Oxidation kinetics and product formation. *Water Res.* 37 (7), 1443–1467.
- von Gunten, U., Elovitz, M., Kaiser, H.P., 1997. Characterization of ozonation processes with conservative and reactive tracers: prediction of the degradation of micropollutants. *Analisis* 25 (7), 29–31.
- von Gunten, U., Elovitz, M., Kaiser, H.P., 1999. Calibration of full-scale ozonation systems with conservative and reactive tracers. *J. Water Supply Res. Technol. AQUA* 48 (6), 250–256.
- von Gunten, U., Salhi, E., Schmidt, C.K., Arnold, W.A., 2010. Kinetics and mechanisms of N-Nitrosodimethylamine formation upon ozonation of N, N-Dimethylsulfamide-Containing Waters: bromide catalysis. *Environ. Sci. Technol.* 44 (15), 5762–5768.
- von Gunten, U., Salhi, E., 2003. Bromate in drinking water a problem in Switzerland? *Ozone Sci. Eng.* 25 (3), 159–166.
- Hammes, F.A., Egli, T., 2005. New method for assimilable organic carbon determination using flow-cytometric enumeration and a natural microbial consortium as inoculum. *Environ. Sci. Technol.* 39 (9), 3289–3294.
- Hammes, F., Salhi, E., Koster, O., Kaiser, H.P., Egli, T., von Gunten, U., 2006. Mechanistic and kinetic evaluation of organic disinfection by-product and assimilable organic carbon (AOC) formation during the ozonation of drinking water. *Water Res.* 40 (12), 2275–2286.
- Hammes, F., Berney, M., Wang, Y.Y., Vital, M., Koster, O., Egli, T., 2008. Flow-cytometric total bacterial cell counts as a descriptive microbiological parameter for drinking water treatment processes. *Water Res.* 42 (1–2), 269–277.
- Helbing, J., Gresch, M., von Gunten, U. Fluorescein as an easy to use tracer for analysis of treatment trains, in preparation.
- Hinterbuchner, T., 2006. Das Verhalten von Benzotriazolen in Abwasserreinigungsanlagen. Diploma Thesis. Fachhochschule Wels, Wels. http://library.eawag.ch/EAWAG-Publications/openaccess/Diplomarbeiten/CHP/UCHEM/diplomarbeit_thinterbuchner.pdf.
- Hoigné, J., Bader, H., 1983. Rate constants of reactions of ozone with organic and inorganic-compounds in water. 1. Non-dissociating organic-compounds. *Water Res.* 17 (2), 173–183.
- Hoigné, J., Bader, H., 1994. Characterization of water-quality criteria for ozonation processes. 2. Lifetime of added ozone. *Ozone Sci. Eng.* 16 (2), 121–134.

- Hollender, J., Zimmermann, S.G., Koepke, S., Krauss, M., Mc Ardell, C.S., Ort, C., Singer, H., von Gunten, U., Siegrist, H., 2009. Elimination of organic micropollutants in a municipal wastewater treatment plant upgraded with a full-scale post-ozonation followed by sand filtration. *Environ. Sci. Technol.* 43 (20), 7862–7869.
- Huber, M.M., Ternes, T.A., von Gunten, U., 2004. Removal of estrogenic activity and formation of oxidation products during ozonation of 17 alpha-ethinylestradiol. *Environ. Sci. Technol.* 38 (19), 5177–5186.
- Huber, M.M., Göbel, A., Joss, A., Hermann, N., Löffler, D., Mc Ardell, C.S., Ried, A., Siegrist, H., Ternes, T.A., von Gunten, U., 2005. Oxidation of pharmaceuticals during ozonation of municipal wastewater effluents: a pilot study. *Environ. Sci. Technol.* 39 (11), 4290–4299.
- Hutchinson, T.H., Hutchings, H.J., Moore, K.W., 1997. A review of the effects of bromate on aquatic organisms and toxicity of bromate to oyster (*Crassostrea gigas*) embryos. *Ecotoxicol. Environ. Saf.* 38 (3), 238–243.
- Karpel Vel Leitner, N., Roshani, B., 2010. Kinetic of benzotriazole oxidation by ozone and hydroxyl radical. *Water Res.* 44, 2058–2066.
- van der Kooij, D., Hijnen, W.A.M., Kruithof, J.C., 1989. The effects of ozonation, biological filtration and distribution on the concentration of easily assimilable organic carbon (AOC) in drinking water. *Ozone-Sci. Eng.* 11, 297–311.
- Krauss, M., Hollender, J., 2008. Analysis of nitrosamines in wastewater: exploring the trace level quantification capabilities of a hybrid linear ion trap/orbitrap mass spectrometer. *Anal. Chem.* 80 (3), 834–842.
- Lee, Y., Escher, B.I., von Gunten, U., 2008. Efficient removal of estrogenic activity during oxidative treatment of waters containing steroid estrogens. *Environ. Sci. Technol.* 42 (17), 6333–6339.
- Lutze, H., 2005. Ozonung von Benzotriazolen. Bachelor Thesis, Universität Duisburg, Essen.
- Macey, R.I., Oster, G.F., Zahnley, T., 2001. Berkeley Madonna. University of California, Berkeley CA.
- Maskaoui, K., Hibberd, A., Zhou, J.L., 2007. Assessment of the interaction between aquatic colloids and pharmaceuticals facilitated by cross-flow ultrafiltration. *Environ. Sci. Technol.* 41 (23), 8038–8043.
- Naik, D.B., Moorthy, P.N., 1995. Studies on the transient species formed in the pulse-radiolysis of benzotriazole. *Radiat. Phys. Chem.* 46 (3), 353–357.
- Nakada, N., Shinohara, H., Murata, A., Kiri, K., Managaki, S., Sato, N., Takada, H., 2007. Removal of selected pharmaceuticals and personal care products (PPCPs) and endocrine-disrupting chemicals (EDCs) during sand filtration and ozonation at a municipal sewage treatment plant. *Water Res.* 41 (19), 4373–4382.
- Neumann, M.B., Gujer, W., von Gunten, U., 2009. Global sensitivity analysis for model-based prediction of oxidative micropollutant transformation during drinking water treatment. *Water Res.* 43 (4), 997–1004.
- Salhi, E., von Gunten, U., 1999. Simultaneous determination of bromide, bromate and nitrite in low $\mu\text{g l}^{-1}$ levels by ion chromatography without sample pretreatment. *Water Res.* 33 (15), 3239–3244.
- Schmidt, C.K., Brauch, H.-J., 2008. N, N-Dimethylsulfamide as precursor for N-Nitrosodimethylamine (NDMA) formation upon ozonation and its fate during drinking water treatment. *Environ. Sci. Technol.* 42 (17), 6340–6346.
- Stalter, D., Magdeburg, A., Weil, M., Knacker, T., Oehlmann, J., 2010. Toxication or detoxication? In vivo toxicity assessment of ozonation as advanced wastewater treatment with the rainbow trout. *Water Res.* 44 (2), 439–448.
- Ternes, T.A., Meisenheimer, M., McDowell, D., Sacher, F., Brauch, H.J., Gulde, B.H., Preuss, G., Wilme, U., Seibert, N.Z., 2002. Removal of pharmaceuticals during drinking water treatment. *Environ. Sci. Technol.* 36 (17), 3855–3863.
- Ternes, T.A., Stuber, J., Herrmann, N., McDowell, D., Ried, A., Kampmann, M., Teiser, B., 2003. Ozonation: a tool for removal of pharmaceuticals, contrast media and musk fragrances from wastewater? *Water Res.* 37 (8), 1976–1982.
- Wert, E.C., Rosario-Ortiz, F.L., Snyder, S.A., 2009a. Using ultraviolet absorbance and color to assess pharmaceutical oxidation during ozonation of wastewater. *Environ. Sci. Technol.* 43 (13), 4858–4863.
- Wert, E.C., Rosario-Ortiz, F.L., Snyder, S.A., 2009b. Effect of ozone exposure on the oxidation of trace organic contaminants in wastewater. *Water Res.* 43 (4), 1005–1014.
- Westerhoff, P., Yoon, Y., Snyder, S., Wert, E., 2005. Fate of endocrine-disruptor, pharmaceutical, and personal care product chemicals during simulated drinking water treatment processes. *Environ. Sci. Technol.* 39 (17), 6649–6663.
- Yamamoto, H., Nakamura, Y., Moriguchi, S., Nakamura, Y., Honda, Y., Tamura, I., Hirata, Y., Hayashi, A., Sekizawa, J., 2009. Persistence and partitioning of eight selected pharmaceuticals in the aquatic environment: laboratory photolysis, biodegradation, and sorption experiments. *Water Res.* 43 (2), 351–362.
- Yang, L., Chen, Z.L., Shen, J.M., Xu, Z.Z., Liang, H., Tian, J.Y., Ben, Y., Zhai, X., Shi, W.X., Li, G.B., 2009. Reinvestigation of the nitrosamine-formation mechanism during ozonation. *Environ. Sci. Technol.* 43 (13), 5481–5487.
- Zhang, Z.L., Zhou, J.L., 2007. Simultaneous determination of various pharmaceutical compounds in water by solid-phase extraction-liquid chromatography-tandem mass spectrometry. *J. Chromatogr. A* 1154, 205–213.
- Zhao, Y.Y., Boyd, J.M., Woodbeck, M., Andrews, R.C., Qin, F., Hrudey, S.E., Li, X.F., 2008. Formation of N-nitrosamines from eleven disinfection treatments of seven different surface waters. *Environ. Sci. Technol.* 42 (13), 4857–4862.
- Zhou, J.L., Liu, R., Wilding, A., Hibberd, A., 2007. Sorption of selected endocrine disrupting chemicals to different aquatic colloids. *Environ. Sci. Technol.* 41 (1), 206–213.